

Let's get started with...

Logic!

Logic

- Crucial for mathematical reasoning
- Used for designing electronic circuitry
- Logic is a system based on **propositions**.
- A proposition is a statement that is either **true** or **false** (not both).
- We say that the **truth value** of a proposition is either true (T) or false (F).
- Corresponds to **1** and **0** in digital circuits

The Statement/Proposition Game

- “Elephants are bigger than mice.”

Is this a statement? yes

Is this a proposition? yes

What is the truth value
of the proposition? true

The Statement/Proposition Game

- “520 < 111”

Is this a statement? yes

Is this a proposition? yes

What is the truth value
of the proposition? false

The Statement/Proposition Game

- “ $y > 5$ ”

Is this a statement? yes

Is this a proposition? no

Its truth value depends on the value of y ,
but this value is not specified.

We call this type of statement a
propositional function or open sentence.

The Statement/Proposition Game

- “Today is January 1 and $99 < 5$.”

Is this a statement? yes

Is this a proposition? yes

What is the truth value
of the proposition? false

The Statement/Proposition Game

- “Please do not fall asleep.”

Is this a statement? no

It's a request.

Is this a proposition? no

Only statements can be propositions.

The Statement/Proposition Game

- “If elephants were red,
- they could hide in cherry trees.”

Is this a statement? yes

Is this a proposition? yes

What is the truth value
of the proposition? probably false

The Statement/Proposition Game

- “ $x < y$ if and only if $y > x$.”

Is this a statement? yes

Is this a proposition? yes

... because its truth value
does not depend on
specific values of x and y .

What is the truth value
of the proposition? true

Combining Propositions

- As we have seen in the previous examples, one or more propositions can be combined to form a single **compound proposition**.
- We formalize this by denoting propositions with letters such as **p, q, r, s**, and introducing several **logical operators**.

Logical Operators (Connectives)

- We will examine the following logical operators:
 - Negation (NOT)
 - Conjunction (AND)
 - Disjunction (OR)
 - Exclusive or (XOR)
 - Implication (if – then)
 - Biconditional (if and only if)
- Truth tables can be used to show how these operators can combine propositions to compound propositions.

Negation (NOT)

- Unary Operator, Symbol: $\neg$

P	$\neg P$
true (T)	false (F)
false (F)	true (T)

Conjunction (AND)

- Binary Operator, Symbol: $\wedge$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction (OR)

- Binary Operator, Symbol: $\vee$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Or (XOR)

- Binary Operator, Symbol: $\oplus$

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication (if - then)

- Binary Operator, Symbol: $\rightarrow$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional (if and only if)

- Binary Operator, Symbol: $\leftrightarrow$

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Statements and Operators

- Statements and operators can be combined in any way to form new statements.

P	Q	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Statements and Operations

- Statements and operators can be combined in any way to form new statements.

P	Q	$P \wedge Q$	$\neg (P \wedge Q)$	$(\neg P) \vee (\neg Q)$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

Equivalent Statements

P	Q	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$	$\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
T	T	F	F	T
T	F	T	T	T
F	T	T	T	T
F	F	T	T	T

- The statements $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$ are **logically equivalent**, since $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$ is always true.

Tautologies and Contradictions

- A tautology is a statement that is always true.
- Examples:
- $R \vee (\neg R)$
- $\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q)$
- If $S \rightarrow T$ is a tautology, we write $S \Rightarrow T$.
- If $S \leftrightarrow T$ is a tautology, we write $S \Leftrightarrow T$.

Tautologies and Contradictions

- A contradiction is a statement that is always
- false.
- Examples:
- $R \wedge (\neg R)$
- $\neg(\neg(P \wedge Q) \leftrightarrow (\neg P) \vee (\neg Q))$
- The negation of any tautology is a contra-
- diction, and the negation of any contradiction is
- a tautology.

Exercises

- We already know the following tautology:
- $\neg(P \wedge Q) \Leftrightarrow (\neg P) \vee (\neg Q)$
- Nice home exercise:
- Show that $\neg(P \vee Q) \Leftrightarrow (\neg P) \wedge (\neg Q)$.
- These two tautologies are known as De Morgan's laws.

Propositional Functions

- Propositional function (open sentence):
- statement involving one or more variables,
- e.g.: $x - 3 > 5$.
- Let us call this propositional function $P(x)$, where P is the **predicate** and x is the **variable**.

What is the truth value of $P(2)$? **false**

What is the truth value of $P(8)$? **false**

What is the truth value of $P(9)$? **true**

Propositional Functions

- Let us consider the propositional function $Q(x, y, z)$ defined as:

- $x + y = z$.

- Here, Q is the **predicate** and x , y , and z are the **variables**.

What is the truth value of $Q(2, 3, 5)$? **true**

What is the truth value of $Q(0, 1, 2)$? **false**

What is the truth value of $Q(9, -9, 0)$? **true**

Universal Quantification

- Let $P(x)$ be a propositional function.
- **Universally quantified sentence:**
- For all x in the universe of discourse $P(x)$ is true.
- Using the universal quantifier $\forall$:
- $\forall x P(x)$ “for all $x P(x)$ ” or “for every $x P(x)$ ”
- (Note: $\forall x P(x)$ is either true or false, so it is a proposition, not a propositional function.)

Universal Quantification

- Example:
- $S(x)$: x is a AASTU student.
- $G(x)$: x is a genius.
- What does $\forall x (S(x) \rightarrow G(x))$ mean ?
- “If x is a AASTU student, then x is a genius.”
- or
- “All AASTU students are geniuses.”

Existential Quantification

- **Existentially quantified sentence:**
- There exists an x in the universe of discourse for which $P(x)$ is true.
- Using the existential quantifier $\exists$:
- $\exists x P(x)$ “There is an x such that $P(x)$.”
- “There is at least one x such that $P(x)$.”
- (Note: $\exists x P(x)$ is either true or false, so it is a proposition, but no propositional function.)

Existential Quantification

- Example:
- $P(x)$: x is a AASTU professor.
- $G(x)$: x is a genius.
- What does $\exists x (P(x) \wedge G(x))$ mean ?
- “There is an x such that x is a AASTU professor and x is a genius.”
- or
- “At least one AASTU professor is a genius.”

Quantification

- Another example:
- Let the universe of discourse be the real numbers.
- What does $\forall x \exists y (x + y = 320)$ mean ?
- “For every x there exists a y so that $x + y = 320$.”

Is it true?

yes

Is it true for the natural numbers?

no

Disproof by Counterexample

- A counter example to $\forall x P(x)$ is an object c so that $P(c)$ is false.
- Statements such as $\forall x (P(x) \rightarrow Q(x))$ can be disproved by simply providing a counter example.

Statement: "All birds can fly."

Disproved by counterexample: Penguin.

Negation

- $\neg(\forall x P(x))$ is logically equivalent to $\exists x (\neg P(x))$.
- $\neg(\exists x P(x))$ is logically equivalent to $\forall x (\neg P(x))$.

Logical Equivalences

- Identity Laws: $p \wedge \mathbf{T} \Leftrightarrow p$ and $p \vee \mathbf{F} \Leftrightarrow p$.
- Domination Laws: $p \vee \mathbf{T} \Leftrightarrow \mathbf{T}$ and $p \wedge \mathbf{F} \Leftrightarrow \mathbf{F}$.
- Idempotent Laws: $p \wedge p \Leftrightarrow p$ and $p \vee p \Leftrightarrow p$.
- Double Negation Law: $\neg(\neg p) \Leftrightarrow p$.
- Commutative Laws:
 - $(p \vee q) \Leftrightarrow (q \vee p)$ and $(p \wedge q) \Leftrightarrow (q \wedge p)$.
- Associative Laws: $(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$
 - and $(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$.

Logical Equivalences

- Distributive Laws:
 - $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$ and
 - $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$.
- DeMorgan's Laws:
 - $\neg(p \wedge q) \Leftrightarrow (\neg p \vee \neg q)$ and
 - $\neg(p \vee q) \Leftrightarrow (\neg p \wedge \neg q)$.
- Absorption Laws:
 - $p \vee (p \wedge q) \Leftrightarrow p$ and $p \wedge (p \vee q) \Leftrightarrow p$.
- Negation Laws: $p \vee \neg p \Leftrightarrow \mathbf{T}$ and $p \wedge \neg p \Leftrightarrow \mathbf{F}$.

Examples

- Find the truth table of $\neg[p \rightarrow (q \oplus r)]$.
- (An important Theorem) Show that:
 - $p \rightarrow q \Leftrightarrow \neg p \vee q$.
- Show the Corollary: $\neg(p \rightarrow q) \Leftrightarrow p \wedge \neg q$.
- Verify the Absorption Laws:
 - $p \wedge (p \vee q) \Leftrightarrow p$, and
 - $p \vee (p \wedge q) \Leftrightarrow p$.

Examples

- Negate:
- (a) For each integer, n , if 4 divides n , then 2 divides n .
- (b) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \ni [x \leq y] \wedge [y \leq (x + 1)]$.